Semester Project | Interactive 2D Potential-Flow Simulator

Senior-level Mechanical Engineering • Programming language: Python, MATLAB, Julia, or equivalent

Objective

Design a graphical program that builds two-dimensional, incompressible, irrotational flows by superposing user-selected elementary solutions. The program shall immediately recompute and display the resulting streamlines, with arrows indicating the local flow direction, whenever the configuration changes.

Required elementary flows and inputs

- Uniform flow — speed U and direction α (measured from the +x axis).
- Source / sink — signed strength m (positive = source; negative = sink) and position (x_0, y_0).
- Point vortex — signed circulation Γ and position (x_0, y_0).
- Sector (wedge) flow — strengths/scaling constant, vertex (x_0, y_0), first-wall angle θ_1 and second-wall angle θ_2; their difference sets the opening angle β.
- Doublet — signed strength μ, location (x_0, y_0) and, if supported, orientation angle.
- Straight symmetry wall — selectable location and orientation; impose zero normal velocity using an appropriate image-system or mathematically equivalent boundary treatment.

User interface and interaction

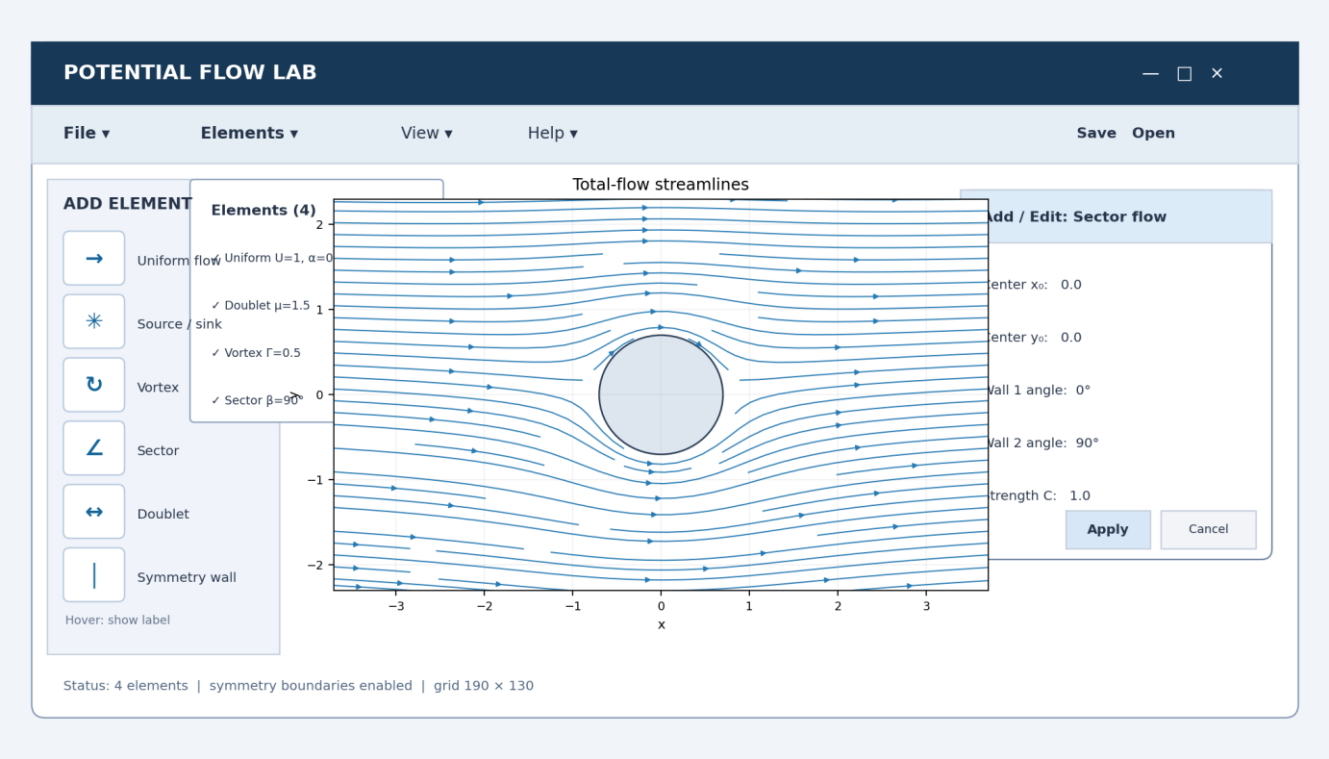
- Left-side icon toolbar: one icon per flow/wall type; hovering reveals its name. Selecting an icon opens a compact parameter-entry dialog with Apply and Cancel.
- Main (x,y) plotting region: show a labeled coordinate system and recomputed streamlines with directional arrows after every add, edit or delete operation.
- Top pull-down "Elements" menu: display all currently added elements and their parameters; allow a selected element to be edited or deleted at any time.
- Top "File" menu: save the complete configuration (types, parameters, walls and plot settings) and reopen it later; include New / Clear All.
- Provide sensible plot limits, streamline density controls and clear handling of invalid parameters or coincident singularities.

Implementation and physical consistency

Superpose velocity components (or potentials where valid), not streamline images. Avoid evaluating singular velocities at the source, vortex and doublet centers. Enforce impermeability for symmetry walls, rather than merely drawing lines. For sector flow, define its admissible wedge domain and state explicitly how its boundaries interact with other elements; unrestricted superposition does not automatically preserve all wall conditions.

Example Interface and Verification Guidance

Illustrative layout (not a mandatory visual design). The pull-down Elements list and the parameter dialog are shown open simultaneously to explain the intended controls.



Minimum verification cases

- Uniform flow at α = 0° and 45°; confirm velocity direction and streamline orientation.
- Single source, sink and vortex: verify sign conventions and expected radial/circular streamlines.
- Uniform flow + doublet: compare with the classical circular-cylinder solution; optional vortex adds circulation.
- Symmetry wall: verify numerically that the wall-normal velocity is zero at representative points.
- Save a configuration, reopen it, and confirm that its element list and streamline plot are reproduced.

Deliverables and assessment

Submit executable code, a user guide, a 4–6 page report (equations, architecture, validation and examples), and a live demonstration. Suggested grading: mathematical correctness 35%; functionality and GUI 35%; verification 20%; documentation 10%.

Recommended optional extensions

Show equipotential lines and velocity-magnitude contours; support zoom/pan, selectable element markers, and exports of images or data. The pressure coefficient may be plotted where a physically appropriate reference velocity and Bernoulli assumptions exist.